Safari: An AI-Driven Sri Lankan Tour Guide Booking Platform

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***Abstract*—Tourism is a domain that plays a critical role in the overall Sri Lankan economy. However, the existing digital platforms for travel planning and tour guide reservation contain several significant limitations, such as inadequate user verification mechanisms, limited personalization, and insufficient safety integrations. To address these drawbacks, this research proposes an AI-powered computerized platform for tour guide booking that enhances trust, privacy, and intelligent matching through an LLM-based agent architecture. The "Safari" system facilitates tourists and guides to make tour requests and respond to such requests, respectively, through voice or text mediums. During these interactions, user privacy is preserved through a detail-locking mechanism that restricts sensitive information until the booking is confirmed. Given the security concerns that might arise, a multi-layered document verification is also set in place to ensure guide authenticity through government-issued documentation and police clearance checks. The developed AI-agent and digital platform are integrated with Google Gemini to perform contextual itinerary recommendations, provide bid-assisted guide matching, and provide emergency guidance. The system is developed by adhering to the Agile methodology practices and is evaluated manually through booking completion rate and agent response accuracy, demonstrating improved reliability and usability for digital tourism services in the context of Sri Lankan tourism services.**

Keywords—AI agents, AI-assisted tourism technology, booking platforms, tourism

I. Introduction

Throughout the times, tourism remains an important sector within Sri Lanka, given the massive contribution it brings to national income, employment generation, and regional development. Especially a country like Sri Lanka that offers diverse attractions, ranging from coastal attractions and biodiversity hotspots to cultural heritage sites, naturally attracts a large number of both international and domestic visitors each year. As a tourism destination, it competes successfully with other destinations partly because of its pivotal geographical position [1]. Consequently, considering how digital transformation is reshaping the global tourism industry, travelers are increasingly relying on online platforms to plan itineraries, discover local experiences of their choice, and connect with service providers such as tour guides. However, despite the availability of widely used international booking platforms, several structural and technological limitations still remain unresolved in the case of Sri Lankan tourism services.

Existing tour booking platforms primarily focus on listing all guides, displaying reviews, and enabling direct contact regardless of their availability or preferences. Therefore, even though these systems provide basic functionality for booking tour guides, they often lack adequate verification processes necessary to ensure the authenticity of service providers. Thus, inadequate verification can expose the tourists to potential risks, including but not limited to engagement with unlicensed or unverified guides, misleading profile information, and possible safety concerns. Furthermore, these platforms generally rely on static filtering methods like price range, ratings, and predefined categories without incorporating any dynamic user preferences, contextual information, or intelligent personalization to match the use case. As a result of this, tourists may experience difficulty in identifying guides who align with their specific requirements, including the budget, group size, itinerary type, and language preferences.

Another critical limitation observed in existing digital solutions is the absence of a privacy-aware workflow, as many conventional systems share contact details and personal information early in the booking process. This increases the risk of personal data misuse or unsolicited communication. In addition, it is observed that emergency handling mechanisms are often limited or external to the platform itself and require the user to rely on national hotlines or third-party applications. These gaps are evidence that highlight the need for a more integrated, secure, and intelligent digital platform tailored to the operational realities in the Sri Lankan context.

Therefore, this research proposes an AI-powered tour guide booking platform, named "Safari", to address these challenges. On a global scale, AI has started actively playing a huge role in creating personalized offers for travelers. These tailor-made solutions make travelling more cost-effective; ensure that the reservations are made with regard to personal preferences; and, in short, they enable the organization of the perfect trip, based on the traveler's demands [2]. Therefore, Safari is also designed specifically to make the tour planning and guide reservation processes convenient, more streamlined, and secure, with enhanced privacy and AI-assistance. The system offers a structured, role-based architecture consisting of three primary user types: tourists, guides, and platform administrators. Tourists can create tour requests either manually through structured forms or via conversational interactions using voice or text. On the other hand, the guides, once their profiles are verified by the administrators, can view relevant tour requests and submit bids either manually or with AI-assistance. Administrators generally oversee the tour guide verification, emergency alerts, and overall platform governance.

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In essence, Safari, as a digital tour assistance platform, aims to improve the convenience, trust, personalization, and operational efficiency in Sri Lanka's tourism landscape by combining structured tour guide booking workflows, multi-step guide verification, conversational AI agents, and emergency support into a single platform.

From the perspective of the research domain, this study contributes to the intersection of AI-assisted service platforms, tourism technology, and digital trust systems. Taking a step beyond the conventional tourism platforms that depend heavily on listing-based discovery and static content, this study introduces an intelligent interaction layer that assists users during requirement elicitation for tour planning and guide matching. The core integration of conversational AI, structured verification mechanisms, and privacy-aware booking workflows highlights how modern AI systems can be utilized to improve reliability and personalization aspects of tourism service platforms in the developing digital ecosystem of Sri Lanka.

II. Literature Review

A. Background of the Sri Lankan Tourism Industry

Sri Lanka is one of the major tourist attraction destinations in the South Asian region. The country has unique opportunities for various kinds of tourism, including traditional tourism, adventure tourism, coastal tourism, eco tourism, safari tourism, cruise tourism, and medical tourism, etc. Its geographical location, natural beauty, and historical and heritage values have made additional potential for promoting tourism in the country [3]. This strategic location and uniqueness have made Sri Lanka a unique tourist destination for centuries. At the end of the 13th Century A.D., Marco Polo visited Sri Lanka, then known as Ceylon, and noted "the traveler reaches Ceylon, which is the most untouchably finest island of its size all the World" [4].

Since the end of the war, Sri Lankan tourism continued to record a two-digit growth rate both in international tourists' arrivals and tourism revenues. This speed growth was disrupted following the Easter Sunday Attack in 2019, which made the industry suffer a loss of US $1.5 billion of tourism revenue, and also the Covid-19 pandemic, which resulted in major losses [5]. Yet, even following such events, the uniqueness of Sri Lanka as a popular tourist destination remains true, with the number of foreigners visiting the country going up year by year to this date.

When national income is concerned, the travel & tourism industry is the third largest export earner in the Sri Lankan Economy, after remittances, textiles, and garments [6]. Working on the front line of tourism reception, tour guides are leading players in enhancing the image and the reputation of tourist destinations in Sri Lanka. The service quality that they deliver is regarded as an important barometer for measuring the overall success of the tourism industry [7].

B. Digital Platforms and Artificial Intelligence in Tourism

In the new digital era, AI works in tandem with human feelings and intelligence, offering customers convenient interaction both online and offline. This approach leads to increased efficiency, productivity, and a better understanding of services [8]. Consequently, AI is one of the potentially most revolutionary and innovative technologies in the tourism industry. AI can contribute to a new digital transformation, changing structures, practices, and how firms collaborate and create value, representing a major technological shift that will affect consumer behavior and impact the industry structure [9].

Yet, in addition to the benefits, there also remain prominent ethical dilemmas and data privacy concerns associated with AI technologies in tourism. Tourists tend to extend their apprehension regarding the collection and use of personal data, highlighting the urgent need for transparent data practices and ethical considerations in deploying AI solutions. These concerns also align with ongoing global discussions about consumer rights and data protection [10].

Recent research in intelligent tourism systems also explores the use of conversational agents and recommendation algorithms to improve the travel planning experiences of tourists. By utilizing natural language processing and machine learning techniques, AI-based tourism assistants are increasingly being used to provide itinerary recommendations, automated customer support, and personalized service discovery. For instance, conversational tourism agents have indicated swift progress in reducing user search effort while improving satisfaction through interactive requirement elicitation and contextual recommendations. Furthermore, integrating large language models and knowledge retrieval mechanisms allows tourism systems to generate more adaptive and context-aware responses compared to traditional rule-based chatbots. These developments highlight the growing role of AI-driven interaction systems in the case of digitizing tourism platforms.

C. Trust, Verification, and Safety in Tourism

Tourism is generally associated with pleasure and leisure activities, and risk is mostly seen as something to be avoided, or perhaps simply as lurking beneath the surface, an ever-present potential threat [11]. Governments, travel agents, and the news media periodically issue warnings about the risks associated with international tourism. Tourists are urged to buy traveler's checks, guidebooks, and bottled water, and to obtain vaccinations as precautions against such risks [12].

Digitalization of the tourism-related workflows introduces new risks expanding into the technological and transaction domains, also. Online platforms can expose tourists to new forms of uncertainty, including identity misrepresentation, fraudulent service listings, privacy breaches, and communication misuse. Similarly, tour guides may also face reputational risks when interacting with anonymous or unverified clients.

D. Comparative Analysis of Existing Digital Tourism Platforms

There are several globally recognized platforms, such as TripAdvisor, GetYourGuide, and Viator, already in the market, catering to requirements such as online tour booking and review-based discovery. But the primary focus of these platforms is to serve as marketplace platforms that support a set of limited features, such as booking and discovering service providers. Even though they provide structured listing and customer feedback mechanisms, their operational models heavily depend on static filtering methods and reputation-based trust systems.

For instance, even though booking functionality and reviews are supported across all three existing solutions stated above, dynamic AI-assisted tour planning that

specifically caters to individual users' preferences is not found in any of them. TripAdvisor provides tour plan matching capabilities, but it is based on a static set of criteria. And most importantly, structured multi-document verification at the individual guide level is not prominently implemented.

In contrast to these solutions, the proposed "Safari" platform uniquely integrates AI-assisted requirement elicitation and travel planning, multi-step guide verification, privacy-locking mechanisms for better safety over sensitive information, and embeds emergency handling within a unified system. It offers a safety-focused design tailored to the Sri Lankan tourism context, all the while providing a more personalized experience.

E. Conclusion

The literature that has been reviewed highlights three interconnected dimensions shaping the modern Sri Lankan tourism landscape: the economic significance, digital transformation, and evolving risk considerations. Sri Lanka's tourism industry plays a critical role in the national growth, and the tour guides serve as the frontline representatives who directly influence visitor experience and destination reputation. AI-based solutions are emerging all over the globe as solutions capable of bridging the gap between tourists and guides, while enhancing the efficiency of the tourism workflows. However, these advancements also come with concerns regarding data privacy, ethical AI deployment, and identity verification.

III. METHODOLOGY

A. Development Approach and Methodology

The development of the "Safari" AI-powered tour guide booking platform has followed the Agile software development methodology. An iterative and incremental approach was adopted to ensure that the functional modules were designed, developed, and validated based on feature priority in progressive cycles. The research process began with requirement analysis through the limitation identification of existing digital tourism platforms, particularly in relation to personalization, offered functionalities, and guide verification.

The system requirements were then categorized according to three primary user roles of tourist, guide, and administrator. For each role, functional workflows were defined, including self-registration, tour request posting, bid placement, booking confirmation, and emergency reporting. Modular components were incrementally implemented and tested based on these workflows. Following the agile approach enabled progressive refinement of the AI interaction flows, early identification of issues, and iterative enhancement of booking and bidding mechanisms. Comprehensive unit and integration testing were conducted during each iteration, including API-level validation and workflow verification to ensure that the system operated as intended.

B. System Architecture Design

Safari platform follows a modular, API-driven client-server architecture that separates presentation logic from backend service operations and integrates AI-assisted orchestration within a controlled execution framework.

The system supports three primary actors, namely the tourists (create tour requests manually or through the AI agent), guides (view tour requests and submit bids manually or with AI assistance), and administrators (approve guide accounts and monitor for emergency alerts). The architectural workflow associated with these three roles can be summarized as:

- A user submits a request through the frontend (text or voice).
- The AI agent processes the request using structured prompts.
- If required, the AI invokes backend API endpoints (example: posting a tour).
- The backend service layer processes the request using a controller–service–repository pattern.
- Data is stored in Firebase Firestore.
- Administrative approval is applied where verification is required.

Key architectural characteristics include a clear separation between frontend and backend layers, endpoint → service → repository interaction pattern, Redis-based session management, and Stripe-based payment integration (Fig. 1).

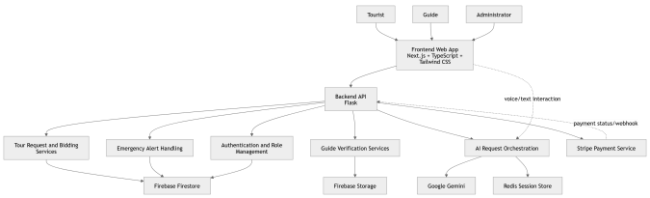

Fig. 1. High-level architecture design

This architecture ensures that AI-driven interactions are adequately controlled and that business logic is executed strictly through validated and dedicated backend services.

C. Technology Stack

The tech stack used in this system development is a modern web technology stack aligned with scalability and modularity principles.

TABLE I. FRONTEND TECHNOLOGIES

Framework	Next.js
Language	TypeScript
Styling	Tailwind CSS

The frontend manages all the user interactions, role-based navigation, and API communication with the backend.

Next.js was selected given its structured routing system, server-side rendering capabilities, and strong ecosystem support that improve performance and maintainability of the frontend application. TypeScript has been chosen to enhance type safety over JavaScript and reduce runtime errors, especially when it comes to a role-based system that involves developing multiple user flows. Tailwind CSS was utilized given its utility-first approach that enables rapid UI development and consistent responsive design across the devices.

TABLE II. BACKEND TECHNOLOGIES

Language	Python
Framework	Flask
Database	Firebase
AI Agent	AI agent powered by a RAG-based knowledge base
Documentation	Swagger
Session Management	Redis

The backend handles authentication, booking workflows, verification processes, emergency alerts, and integration with external services.

Python was selected as a result of its strong ecosystem support for AI integration and rapid backend development. Flask, which is a lightweight framework, was used as it enables modular API construction and flexible integration with AI services like Google Gemini. Firebase was selected for its managed cloud-based infrastructure, real-time data handling, and secure storage capabilities. Swagger was used to standardize the API documentation. Redis was utilized for session management because of its in-memory performance efficiency, which enables fast state tracking for AI-driven multi-step interactions.

As for the AI integration, Google Gemini is the foundational large language model used here. Google Gemini has advanced natural language understanding capabilities and the ability to support structured prompt-based interactions with high accuracy without custom model training in this context. Payment processing is performed through a Stripe integration.

D. AI Agent Process Workflow

The AI component of the Safari platform is powered by Google Gemini. The system integrated Gemini through structured prompts and controlled API invocation mechanisms instead of training a proprietary model.

The AI agent is capable of performing requirement elicitation through conversational interaction, identification of missing booking details, contextual itinerary suggestions, bid-assistance for guides, and emergency aid (Fig. 2).

E. Verification and Emergency Alert Handling Workflows

Another core component of the methodology involves ensuring the security and safety of the tourists within the platform. For this, it is essential that the guides complete a three-step Know Your Customer (KYC) process before account activation, where the administrator approves their NIC, a photo of the tourist board-issued license, and a police clearance report. Only after that is a guide permitted to access their account to apply for tour requests.

Tourists are also expected to submit a photo of their NIC or passport for verification purposes to prevent fraudulent activities and ensure the users' safety. Furthermore, during an emergency that occurs in an active booking, the tourists can trigger an emergency alert that immediately gets sent over to the admin portal for the administrator's review.

IV. Results and Analysis

The implemented system has been evaluated based on functional performance, AI interaction effectiveness, and overall workflow integrity with particular attention to AI-assisted booking lifecycle and guide application mechanisms.

A total of 10 structured functional test scenarios were taken as the foundation to perform testing, based on the system requirements initially identified during requirement gathering. These scenarios cover the end-to-end interaction flows, including AI-driven trip planning, booking submission,

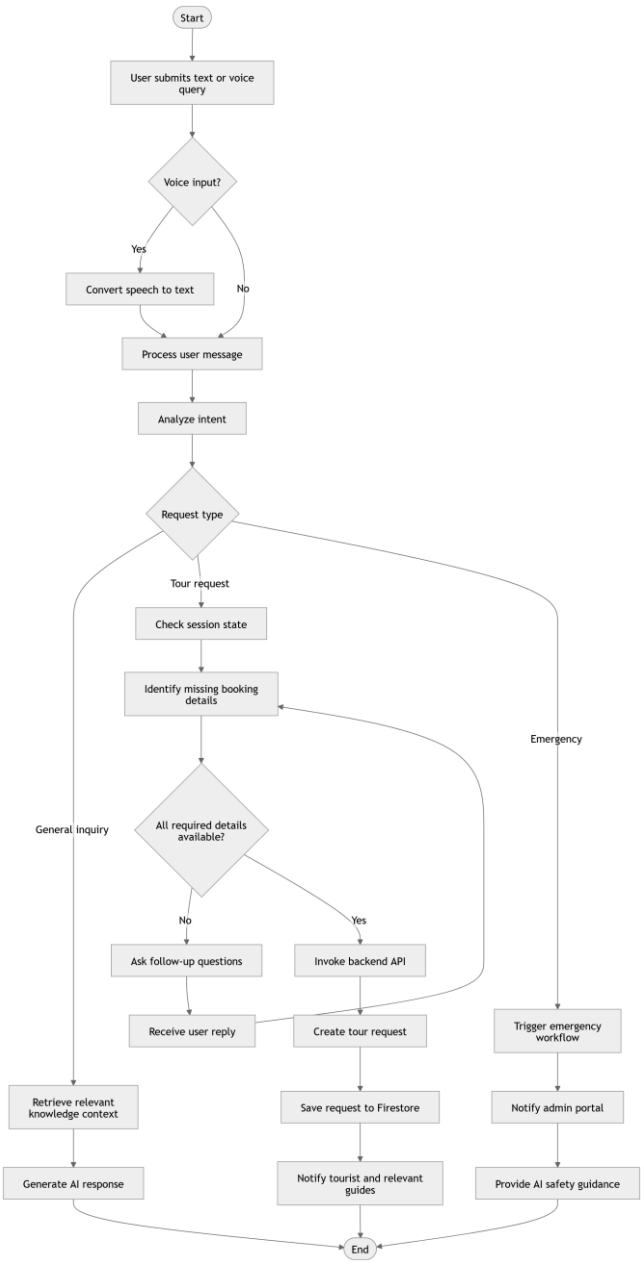

Fig. 2. System AI interaction workflow for general inquiries, tour requests, and emergency handling

guide bid handling, booking confirmation, privacy unlocking, and emergency features.

Taken as a total, 15 complete booking flow cycles were manually executed during controlled testing, in addition to API and unit testing. Each booking flow included the stages of:

1. Tourist trip request creation (voice or text in English, Sinhala, or Tamil languages).

2. AI parameter elicitation.

3. Tour request posting via backend API.

4. Guide review and bid submission.

5. Tourist confirmation and booking finalization.

6. Privacy detail unlocking only after booking confirmation.

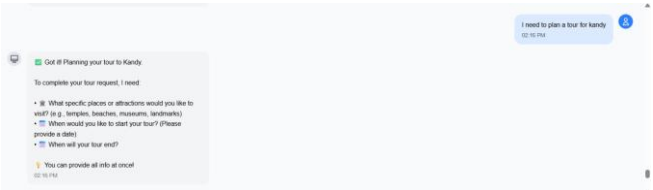

Fig. 3. Gemini requesting further information to create a tour request

For instance, when a tourist user creates and posts a tour request using the AI agent, Gemini guides the user by sequentially collecting the required parameters (Fig. 3). If necessary information is missing, the agent is designed to prompt for clarification. Once all required criteria are fulfilled, the agent triggers the appropriate backend API to post the tour request. This structured iteration for all three essential details reduces the risk of incomplete submissions and improves booking workflow consistency.

Similarly, guide users can utilize the AI assistance when filtering available tour requests and placing bids. The system also incorporates a Retrieval-Augmented Generation (RAG)-based knowledge mechanism to provide timely and contextual responses. Furthermore, voice interaction is also supported via speech-to-text conversion, enabling tourists to interact with the AI agent through spoken queries.

Evaluation parameters taken for quality assurance were booking completion rate and AI response accuracy based on the provided knowledge base and other data. During the controlled testing, all 15 of the booking flows were successfully completed without any major workflow interruptions. Only minor gaps were observed during AI parameter extraction. Across these tested scenarios, the AI agent correctly extracted relevant parameters in approximately 12 out of 15 interactions (80%), while the remaining cases required additional clarification prompts from the system. API execution success rate during booking submission and bid placement was observed as 97% across all tested scenarios and flows, demonstrating stable backend interaction during the evaluations. These observations from the results indicated improved efficiency, consistent data capture, and reliable API execution, demonstrating the practical feasibility of integrating structured AI within a digital tour guide booking platform.

V. Conclusion

As Sri Lanka continues to grow as a significant tourist destination in South Asia, the growing reliance on digital systems for traveling and booking services requires more secure and intelligent mechanisms. Thus, this research addressed major limitations observed in existing digital tourism-related platforms within the Sri Lankan context, specifically in relation to personalization, convenience, verification, and trust. Conventional platforms existing today mainly depend on static filtering and reputation-based trust models, which are not sufficient to mitigate issues such as misrepresentation, privacy exposure, or transactional uncertainty. As a response to these challenges, this study proposed and developed "Safari", an AI-powered tour guide booking platform that combines structured verification workflows and privacy-aware booking mechanisms that cater to the users' requirements. A key contribution of the system lies in the integration of the Google Gemini-based AI agent capable of structured requirement elicitation and controlled backend API invocation. The agent assists tourists in completing tour requests and supports guides in filtering requests and applying for requests. Apart from the above, the incorporation of RAG-based knowledge mechanisms has enhanced the contextual accuracy, while speech-to-text capability has improved the accessibility through voice interaction.

Evaluation through controlled manual testing has demonstrated consistent booking workflow completion, reliable API execution, and AI-guided interaction, fulfilling the aim of the research to provide a comprehensive digital solution that addresses the gaps between tourist-guide interactions in existing systems. Future enhancements may include broader testing with targeted users, advanced fraud-detection analytics, integration with real-time external data sources for expanded functionalities, and extending into mobile application environments. All in all, the proposed system contributes to advancing secure, intelligent, and context-aware tourism technology with Sri Lanka's evolving digital ecosystem.

References

[1] S. Fernando, J. S. Bandara , and C. Smith, "Tourism in Sri Lanka - The Routledge Handbook of Tourism in Asia," 2017, Available: https://www.academia.edu/download/51847037/Tourism_in_Sri_Lanka_-_The_Routledge_Handbook_of_Tourism_in_Asia.pdf

[2] M. Zsarnoczky, "HOW DOES ARTIFICIAL INTELLIGENCE AFFECT THE TOURISM INDUSTRY?," Journal of Management, vol. 31, no. 2, 2017.

[3] P. M. B. Jayathilake, "TOURISM AND ECONOMIC GROWTH IN SRI LANKA: EVIDENCE FROM COINTEGRATION AND CAUSALITY ANALYSIS," International Journal of Business, Economics and Law, vol. 2, no. 2, Jun. 2013.

[4] S. Fernando, J. S. Bandara, and C. Smith, "Tourism in Sri Lanka," 2017.

[5] W. Samarathunga, "Post-COVID19 Challenges and Way Forward for Sri Lanka Tourism," Apr. 2020, Accessed: Feb. 14, 2026. [Online]. Available: https://ssrn.com/abstract=3581509

[6] L. M. Munasinghe, T. Gunawardhana, and R. G. Ariyawansa, "Sri Lankan Travel & Tourism Industry: Recent Trends and Future Outlook towards Real Estate Development," presented at the 2018 International Conference On Business Innovation (ICOBI), Aug. 2018.

[7] J. A. R. C. Sandaruwani and W. K. A. C. Gnanapala, "THE ROLE OF TOURIST GUIDES AND THEIR IMPACTS ON SUSTAINABLE TOURISM DEVELOPMENT: A CRITIQUE ON SRI LANKA," Tourism, Leisure and Global Change, vol. 3, 2016.

[8] A. N. Kazak, P. V. Chetyrbok , and N. N. Oleinikov, "Artificial intelligence in the tourism sphere," IOP Conference Series: Earth and Environment, 2020, Available: https://iopscience.iop.org/article/10.1088/1755-1315/421/4/042020

[9] R. Filieri, E. D'Amico, A. Destefanis, E. Paolucci, and E. Raguseo, "Artificial intelligence (AI) for tourism: an European-based study on successful AI tourism start-ups," Oct. 2021, Available: https://www.emerald.com/ijchm/article-abstract/33/11/4099/126338/Artificial-intelligence-AI-for-tourism-an-European?redirectedFrom=fulltext

[10] A. Ferhataj and F. Memaj, "CHALLENGES AND OPPORTUNITIES OF AI IMPLEMENTATION IN TOURISM: AN ETHICAL AND TECHNOLOGICAL PERSPECTIVE," 2024, doi: https://doi.org/10.52320/svv.v1iIX.357.

[11] A. Williams and V. Baláž, "TOURISM RISK AND UNCERTAINTY: THEORETICAL REFLECTIONS," Journal of Travel Research, 2015, Available: https://journals.sagepub.com/doi/abs/10.1177/0047287514523334

[12] A. Lepp and H. Gibson, "TOURIST ROLES, PERCEIVED RISK AND INTERNATIONAL TOURISM," Annals of Tourism Research, vol. 30, 2003, doi: https://doi.org/10.1016/S0160-7383(03)00024-0.